Where Does Long-Term Agent Memory Actually Fail? A Lifecycle-Boundary Attribution Protocol with Oracle Controls

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Abstract

Long-term-memory agents are evaluated end-to-end: a wrong answer is scored wrong, and nothing is learned about where the pipeline broke. We present Lifecycle-Boundary Attribution, a frozen protocol that attributes every wrong answer to a specific boundary of a nine-stage memory lifecycle (conversation → formation → storage → evolution → retrieval → reranking → context → reasoning → answer). Three design choices make the attribution reliable: an anti-circular evidence-sufficiency gate that compares gold evidence against the benchmark reference answer rather than the model's own prediction; a semantic decision layer in which lexical metrics never enter attribution; and two oracle controls—gold injected as formed memories into an independent engine vs. directly into the context—that separate retrieval-side from reasoning-side failures. Running the frozen protocol on two benchmarks (LongMemEval-s and LoCoMo), two answer models, two formation extractors, and an external open-source memory system (Mem0), with paired statistics and an adjudicated judge-reliability study, we find that the dominant boundary is formation, not retrieval: real memory loses a large fraction of gold evidence before it ever reaches the store. The boundary is a property of the extractor, not of the pipeline—swapping the formation extractor inside one fixed pipeline collapses the oracle-memory advantage from +17.3 points to zero—and it shifts downstream on long conversations; the same label structure and gate criticality reproduce on Mem0. An attribution protocol rather than a single bottleneck claim is the transferable contribution. The protocol, the measurement-validity fixes it required (each with regression tests), and the full per-question traces are released as an open artifact.

1 Introduction

Agent memory systems—retrieval-augmented stores, consolidation pipelines, reflective memories—are increasingly the substrate of long-horizon LLM agents, and they are evaluated almost exclusively end-to-end: answer a question over a long conversation, score the answer. That score conflates every way the system can fail. A missing fact, a superseded belief, a retrieval miss, a truncated context window, and a reasoning slip all produce the same observable: a wrong answer.

Prior work on memory failures either taxonomizes errors post-hoc from traces or reports aggregate accuracy, and the attribution instruments underneath carry three measurement problems that we document and fix here: (i) lexical scorers (exact match, token F1) are used as ground truth for correctness, conflating representation failures with reasoning failures; (ii) evidence matches by lineage (a fact citing a source turn) are treated as content preservation; (iii) "oracle" controls are underspecified—injecting gold evidence into the context and injecting it into memory are different interventions, and equating them misattributes retrieval failures to reasoning. Previewing the answer: the bottleneck sits in the formation-stage implementation, not the pipeline—and which stage binds moves with extractor, retriever, and workload.

Contributions.

1. A frozen attribution protocol with an anti-circular evidence gate. Lifecycle-Boundary Attribution labels every wrong answer with the earliest lifecycle boundary at which gold evidence is not fully present. Before any failure label is assigned, an E0_evidence_insufficient gate

asks whether the benchmark's own reference answer is recoverable from the gold evidence—comparing against the reference rather than the model's prediction, which breaks the "wrong answer → missing evidence → blame memory" circularity. All thresholds are frozen before any large-scale run (§3).

2. Measurement-validity fixes, each with regression tests. Lexical scores never enter attribution; provenance matches require content preservation; semantic evidence matching requires faithful representation rather than topical overlap; deterministic judge channels are narrowed to quote/date rules; embedding-backend failures fail loudly instead of silently degrading.

3. A judge-reliability study with author adjudication. We measure inter-judge agreement (Cohen's κ) between two judge models on a seeded subsample, adjudicate every disagreement against gold evidence under published guidelines, validate the adjudication against two further blind judge families, and report how the headline numbers move under the adjudicated verdicts (Study 2E). To our knowledge this is the first adjudicated judge-reliability measurement in long-term memory evaluation.

4. An empirical study on two benchmarks, two answer models, and two formation extractors, plus an external open-source memory system (Mem0), with paired statistics (exact McNemar, paired bootstrap CIs), per-question-type breakdowns, and a cross-model replication. The dominant failure boundary is formation; reasoning failures (F6 candidates) are rare and mostly control failures of the oracle conditions. An extractor ablation (Study 2F) shows the boundary is a property of the formation stage, not of one rule-based implementation: with an LLM-driven extractor, formation recall rises from 0.236 to 0.838 and the oracle-memory advantage collapses from +17.3 points to zero, relocating the residual bottleneck to benchmark evidence sufficiency and reasoning. An external-system replication (Study 3A) shows the failure taxonomy transfers: on Mem0 the label distribution and oracle contrast structure match the in-house run on the same questions.

The three-condition design (Real Memory / Oracle Memory / Oracle Context) is the measurement instrument these contributions run on, not a contribution in itself: oracle-style controls have become common in memory evaluation (Long et al., 2026; Sidik and Rokach, 2026). What is new is the protocol around them—the anti-circular gate, the frozen thresholds, the regression-tested validity fixes, and the adjudicated judge study.

2 Related Work

Long-term memory benchmarks. LongMemEval (Wu et al., 2025) evaluates chat assistants over long histories with question types that stress extraction, temporal reasoning, and knowledge updates; LoCoMo (Maharana et al., 2024) evaluates very long multi-session conversations. We use both as evidence-boundary substrates and treat their gold evidence spans as the lifecycle signal.

Memory architectures. RAG-style stores (Lewis et al., 2020; Robertson and Zaragoza, 2009), consolidation and reflection pipelines (Park et al., 2023; Packer et al., 2023; Shinn et al., 2023), and recent agent-memory systems (Chhikara et al., 2025; Xu et al., 2025; Liu et al., 2024b) optimize recall or answer accuracy. They report what the system remembers or answers; ours reports where along the lifecycle the evidence was lost, on a concrete open implementation, so the protocol's claims are checkable rather than asserted.

Failure analysis and attribution. Error taxonomies for agents and RAG failure analyses (Zhang et al., 2024) typically require human trace reading; long-context degradation studies (Liu et al., 2024a) characterize attention-side rather than memory-pipeline loss. Our protocol automates lifecycle attribution with measured judge reliability (§5).

Diagnosing where memory fails (2024–2026). A first wave of automated failure-

attribution work targets multi-agent systems: Who&When localizes the responsible agent and decisive step (Zhang et al., 2025b); AgenTracer labels failure traces by counterfactual replay and fault injection (Zhang et al., 2025a); AgentDebug classifies errors into memory, reflection, planning, action, and system modules (Zhu et al., 2025). Two 2026 preprints probe long-term memory systems specifically, and their conclusions differ from ours instructively. MemTrace measures per knowledge point across 13 configurations and four paradigms and reports the dominant bottleneck as evidence use, not retrieval: when systems fail, the evidence was retrievable ten times more often than it was missing (Long et al., 2026). MEMTIER analyzes retrieval bottlenecks in a tiered architecture on LongMemEval-S with Qwen2.5-7B, reporting that structured pre-population lifts single-session recall to 0.686–0.714 (Sidik and Rokach, 2026). A concurrent LoCoMo diagnostic goes further: Yuan et al. (Yuan et al., 2026) report that retrieval method dominates (a 20-point spread) while write strategy moves accuracy 3–8 points, and that raw chunk storage with zero LLM calls beats expensive extraction. We read the four-way divergence—MemTrace's "evidence use", MEMTIER's and Yuan et al.'s "retrieval", our formation dominance—as the motivation for a protocol rather than a contradiction: the dominant boundary is a property of (write-path quality × retrieval quality × workload shape), so no pooled score can locate it, and the disagreements are predictable from the lifecycle view (§??): raw-storage-wins is the zero-formation-loss limit, retrieval-dominance reflects a wide method spread (or long conversations, our Study 2C), and evidence-use-dominance reflects near-ceiling retrieval. Study 2F isolates the write-path variable inside one fixed pipeline (the oracle-memory advantage collapses from +17.3 points to zero), Study 2C shifts the workload, Study 3A varies the system. Oracle-style controls are now common here (Long et al., 2026; Sidik and Rokach, 2026; Yuan et al., 2026); our claim is the attribution protocol around them, not the oracle construction.

Evaluation methodology. LLM-as-judge and semantic evaluation (Zheng et al., 2023; Liu et al., 2023); judge reliability and self-preference bias, uncertainty and abstention (Kadavath et al., 2022). We measure inter-judge agreement explicitly (Cohen's κ, Study 2E). For condition comparisons we use exact McNemar tests (McNemar, 1947) with paired bootstrap CIs (Efron, 1979), and treat questions within one conversation as correlated (Dietterich, 1998), reporting per-type breakdowns rather than pooling silently.

3 Protocol: Lifecycle-Boundary Attribution

Lifecycle. Conversation → Formation → Storage → Evolution → Retrieval → Reranking → Context → Reasoning → Answer. Each arrow is a measurable boundary with its own artifact set.

Labels. F1_formation (evidence never formed), F2_storage (formed but not stored), F3_evolution (stored but superseded/dropped by governance), F4_retrieval (in store but not retrieved), F5_reranking (retrieved but dropped by reranking), F6_reasoning _candidate (all evidence reached context, answer still wrong), E0_evidence_insufficient (gold evidence insufficient for the reference answer), eval_semantic_uncertain (judge undecided), rc_reasoning_control_failure (oracle control answered wrong).

Decision order. (1) Semantic verdict on the model answer; correct → no failure. (2) Undecided → eval_semantic_uncertain. (3) E0 gate: is the gold evidence sufficient for the benchmark reference answer (never the model prediction)? Insufficient → E0_evidence_insufficient. (4) Earliest lifecycle stage whose artifact set does not fully contain the gold evidence → F1–F5. (5) All stages reached and answer still wrong → F6_reasoning _candidate.

Evidence matching tiers. Exact text → provenance with content-preservation check (shared content vocabulary $\geq$ 50% of the smaller side and $\geq \min(2, |gold|)$ shared tokens; rejected matches are reported as provenance_only) → semantic (faithful-representation containment at 0.8, either

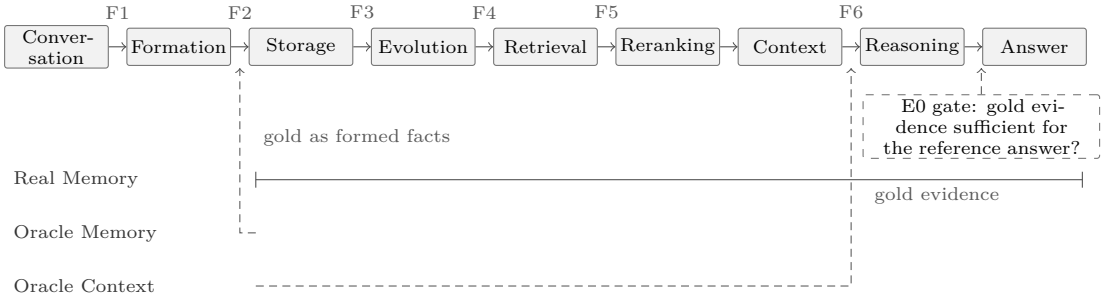


Figure 1: Lifecycle chain with the failure label each boundary names (F1–F6). The E0 gate tests gold-evidence sufficiency for the benchmark reference answer before any F-label is assigned. The three conditions differ only in where gold evidence enters: Real Memory runs the full pipeline; Oracle Memory injects gold as formed facts at the Formation→Storage boundary; Oracle Context injects gold at the Context→Reasoning boundary.

direction, same absolute floor). Raw-token overlap is never a match: two same-topic different-event items ("the park on 9 May 2023" vs "the park on 7 May 2023") score 0.56 under the legacy rule and are conflated, one mechanism behind inflated legacy recall.

Unmeasured stages. Stages a condition does not measure carry measured: false and are skipped, never read as perfect recall. Oracle Context, for example, bypasses the pipeline; its unmeasured stages are excluded from stage recall.

Embedding-backend discipline. Embeddings serve retrieval only; attribution runs on content. A failing embedding request is retried and then fails the record loudly (hash fallback is opt-in offline mode); inputs exceeding the backend context window are deterministically halved; every run is verified dimension-clean afterwards (§4).

4 Experimental Design

System under test. A concrete open implementation: a formation extractor, an embedding store with dense retrieval and reranking, context assembly, and a local answer model (qwen2.5:7b, temperature 0; gemma3:12b cross-check), bge-m3 embeddings. The formation extractor has two arms behind one interface: a deterministic rule-based analyzer (facts, relations, events, rules, profiles) and an LLM-driven extractor (qwen2.5:7b, Study 2F). Everything downstream of formation— governance, storage, retrieval, reranking, context assembly—is identical across arms, so the extractor is the only varying component. The rule arm makes no LLM calls at all; only answer generation, semantic judging, embeddings, and the 2F extraction arm hit the model server.

Conditions. Real Memory: full pipeline over the conversation. Oracle Memory: gold evidence injected as formed active facts into an independent engine, then the full retrieval/reranking/context pipeline. Oracle Context: gold evidence placed directly into the answer context, bypassing the pipeline. The three conditions partition failure responsibility: Real vs. Oracle Memory isolates the memory-side loss; Oracle Memory vs. Oracle Context isolates context assembly; Oracle Context's own wrong answers are rc_reasoning_control_failure.

Datasets and sampling. LongMemEval-s (n=10 full haystack, protocol replication); the LongMemEval oracle subset (a primary n=200 run—the first 200 items of the oracle source—alongside a two-stage stratified n=50 pilot, reused by the controlled ablations 2D/2E/2F); LoCoMo (n=50). Stratification recipes and seeds are in Appendix F; question-level subsets are committed, conversations joined at load time from the local full data, and raw benchmark data is not redistributed.

Metrics and statistics. Per-condition semantic accuracy (judge verdict; lexical EM/F1 reported in parallel, never as attribution input); per-stage recall; failure-label distribution. Condition comparisons use exact McNemar tests on discordant pairs and paired boot-

strap percentile CIs (10,000 resamples). Judge reliability uses Cohen's κ between the primary judge and a second judge model re-judging the same stored answers.

Hypotheses. H1: Real Memory loses a significant fraction of gold evidence before context; F1/F4/F5 dominate over F6 candidates. H2: Both oracle conditions exceed Real Memory while Oracle Context stays below ceiling, so evidence availability dominates the gap and reasoning on perfect evidence remains imperfect. H3: Real-vs-oracle inversions (oracle wrong, real right) are control failures, not evidence of reasoning strength.

5 Results

Study 2A (protocol replication, n=10, full haystack). Three-condition independence holds in the strict-mode rerun: Real Memory 7/10, Oracle Memory 10/10, Oracle Context 10/10; the real-vs-oracle gap is directionally large (+0.300) but not significant at n=10 (McNemar p=0.25, CI $[+0.000, +0.600]$)—this study validates the protocol, not estimates effects. First losses F1_formation×8, F3_evolution×1, context×1; the E0 gate finds all 10 items evidence-sufficient.

Study 2B (primary, n=200). Table 1 reports the scaled run on the LongMemEval oracle subset (n=200; answer and judge model deepseek-v4.1-flash, temperature 0). Semantic accuracy (uncertain verdicts excluded from the denominator) is 52.1% for Real Memory (85/163), 69.1% for Oracle Memory (96/139), and 82.1% for Oracle Context (110/134). Real Memory loses gold evidence at the formation boundary in 155/200 cases (77.5%): formation recall 0.427 (95% CI $[0.375, 0.480]$; bootstrap over questions), against storage/retrieval at 1.000, rerank 0.990, and context 0.896 (CI $[0.868, 0.922]$); evolution recall ties formation mechanically (facts missing at formation are absent downstream). Oracle Memory restores every pre-context stage to 1.0 and reaches 0.951 context recall (CI $[0.932, 0.968]$), its residual losses being context-assembly, not formation. The E0 gate finds 153/200 items evidence-sufficient, 41 insufficient (reported as E0_evidence_insufficient, not memory failures), and 6 uncertain (excluded). Among attributable answer failures, F1_formation dominates (61, 72.6%), followed by E0_evidence_insufficient (10), eval_semantic_uncertain (8), and F6_reasoning (3); only one F4_retrieval and one F5_reranking occur, because gold evidence rarely survives formation. Disabling the gate reclassifies all 10 E0_evidence_insufficient items as lifecycle failures (9 as F1_formation), inflating F1_formation 61→72 (+18%) and the formation share 73%→86%: the gate is load-bearing, not cosmetic.

	Real	Oc. Mem.	Oc. Ctx.
Semantic accuracy*	52.1%	69.1%	82.1%
Lexical accuracy†	31.7%	46.0%	55.0%
Formation recall	0.427	1.000	n/a
Storage recall	1.000	1.000	n/a
Evolution recall	0.427	1.000	n/a
Retrieval recall	1.000	1.000	n/a
Rerank recall	0.990	1.000	n/a
Context recall	0.896	0.951	1.000
First loss = F1	155/200	0/200	n/a

Table 1: Study 2B (LongMemEval oracle subset, n=200). F1 = F1_formation. *Semantic accuracy = 85/163, 96/139, 110/134 (uncertain verdicts excluded from the denominator). †Lexical (token-F1) accuracy = 63/199, 92/200, 110/200, reported for reference only and never an attribution input.

Both oracle contrasts are decisive. Real versus Oracle Memory: McNemar b=20, c=5, p=0.0041; bootstrap 95% CI $[+0.047, +0.195]$. Real versus Oracle Context: b=29, c=5, $p<0.0001$, CI $[+0.111, +0.299]$ (cluster-robust by conversation: identical, since LongMemEval is one question per conversation, 200/200). The oracle-versus-oracle ordering is not robust—Oracle Context exceeds Oracle Memory here and in 2D's gemma arm but not in the n=50 qwen pilot—so we read the oracle contrast as one combined real $\ll$ oracle effect (H2), with Oracle Context below ceiling showing reasoning on perfect evidence is itself imperfect and its edge over Oracle Memory showing the downstream assembly is lossy.

Study 2C (cross-benchmark, LoCoMo n=50). LoCoMo questions are embedded in much longer multi-session conversations (∼600 messages per question across ∼28 sessions, versus ∼115K-token histories in Study 2B), so this

Question type	n	Real	Oracle Mem.	Oracle Ctx.	Attributed failures (real)
knowledge-update	78	0.582	0.688	0.755	F1×14, E0×5, EVAL×4, F6×2
multi-session	62	0.278	0.405	0.788	F1×34, E0×3, EVAL×4, F4×1, F5×1
temporal-reasoning	60	0.704	0.939	0.917	F1×13, E0×2, F6×1

Table 2: Study 2B semantic accuracy by question type (n=200; the subset draws on the three harder LongMemEval types). Failure labels abbreviated: F1 = F1_formation, E0 = E0_evidence_insufficient, EVAL = eval_semantic_uncertain, F4 = F4_retrieval, F5 = F5_reranking, F6 = F6_reasoning. Formation failures concentrate in multi-session items. Per-type cells are descriptive (smallest n=60); the MDE at 80% power is $\Delta \approx +0.23$ (Appendix G).

study tests whether the formation-dominant boundary survives a change of benchmark and workload shape. It does, with one informative shift. Semantic accuracy is 26/50 = 52.0% (real) vs. 36/50 = 72.0% (oracle memory) vs. 37/50 = 74.0% (oracle context); both oracle contrasts are significant (real vs. oracle memory: b=15, c=5, p=0.0414, CI $[+0.040, +0.360]$; real vs. oracle context: b=16, c=5, p=0.0266, CI $[+0.040, +0.380]$). First loss is again concentrated at formation (F1_formation 46/50); the answer-failure distribution is F1_formation×18, E0_evidence_insufficient×3, F4_retrieval×3 (E0 gate: 47 sufficient, 3 insufficient). The shift is on the retrieval side: retrieval recall falls 1.000→0.608 (CI $[0.483, 0.728]$) and reranking 0.990→0.527 (CI $[0.398, 0.655]$) once conversations grow to hundreds of messages, so F4_retrieval losses appear (3/24 attributable). Formation recall is even lower than 2B (0.097, CI $[0.028, 0.182]$): the boundary moves with the workload but stays pre-context.

Lexical false negatives are even stronger on LoCoMo (lexical 28.0% vs. semantic 52.0% on real answers), reinforcing that lexical scoring would misattribute most of these failures.

Study 2D (cross-answer-model control, n=50 subset). On the n=50 pilot subset, swapping the answer model qwen2.5:7b for gemma3:12b (judge fixed) tests which parts of the attribution are properties of the pipeline rather than of the answer model. The lifecycle side is unchanged exactly as the protocol predicts—real-memory stage recall is identical across the two answer models (formation 0.236, first loss F1_formation 45/50), because extraction never invokes the answer model. Only the answer side moves: qwen scored 66.0/83.3/75.0% and gemma compresses to 59.6/65.3/69.4% (real/oracle memory/oracle context), the real-vs-oracle-memory contrast losing significance (b=7, c=4, p=0.549) while real vs. oracle context stays directional (c=0; b=5, p=0.063). As at scale in 2B, the oracle ordering flips under gemma (oracle context > oracle memory); one multi-session inversion appears (real 0.500 vs. oracle memory 0.417). With the judge fixed across the two answer models, formation dominance cannot be an artifact of judge–answer-model co-adaptation.

Study 2E (judge reliability). Gemma3:12b re-judges the stored answers of a seeded 20-record subsample of the n=50 pilot subset (seed 20261018): 20 questions × 3 conditions = 60 answers, 55 with paired verdicts (5 abstentions). Raw agreement is 43/55 = 78.2%, Cohen's $\kappa = 0.158$—a prevalence effect: judges agree on the 41/55 consensus-correct items and disagree only inside a 14-verdict borderline band. We adjudicated every disagreement (12 conflicting pairs plus 4 abstention cases; 16 total, 14 decidable after excluding two preference items): the qwen2.5:7b judge matches adjudicated truth 8/14, gemma3:12b 3/14 (gemma is lenient: 9 over-acceptances, 0 over-rejections; qwen balanced 2/2). All denominators are tabulated in Appendix D; the deterministic rule stack resolved 9/60 verdicts identically for any judge, so it is not a disagreement source.

Blind validation of the adjudication (two external families). The adjudication is itself a measurement, so we validate it against judge families that cannot have co-adapted with either primary judge, re-judging the same 60 answers under the identical prompt and rules, strictly blind (question, answer, and gold evidence only). DeepSeek (deepseek-v4.1-flash, API) reproduces the adjudication on 13/16 cases

category	n	Real	Oracle Mem.	Oracle Ctx.	failures (real)
multi-hop	10	0.500	0.700	0.800	F1×5
temporal	10	0.300	0.600	0.600	E0×3, F1×2, F4×2
open-domain	9	0.333	0.444	0.444	F1×6
single-hop	21	0.714	0.905	0.905	F1×5, F4×1

Table 3: Study 2C semantic accuracy by LoCoMo category (1=multi-hop, 2=temporal, 3=open-domain, 4=single-hop); label abbreviations as in Table 2, F4 = F4_retrieval. Temporal items are weakest; single-hop items, though best, still lose 5/21 to formation. Category cells are small (n=9–21) and exploratory: only effects well above the study-level MDE are interpretable at this granularity.

(13/14 decidable) while agreeing with neither primary judge (9/16 qwen, 4/16 gemma); a fourth family (Mistral mistral:7b, local, 60/60 verdicts) does not corroborate on the borderline band (1/14 with adjudicated truth, siding with the lenient gemma judge 8/14). The four families sort by leniency rather than noise (Appendix D): subsample acceptance 40% < 74% < 90% ≈ 90%, borderline-band agreement with adjudicated truth in the same order (13/16 > 8/14 > 3/14 > 1/14), so the adjudicated verdicts track the strictest gold-evidence reading; on the 41-item consensus core all families agree (mistral 41/41). Applying the adjudicated verdicts moves the n=50 pilot's numbers at most 4.2 points (real 66.0%→62.5%, Oracle Memory 83.3%→81.6%, Oracle Context 75.0%→79.2%), leaves the ordering unchanged, and strengthens the primary contrast—the lifecycle attribution is robust to judge choice.

Study 2F (extractor ablation, n=50). The decisive control: the same 50 LongMemEval-s oracle questions, answer model, judge, embeddings, and lifecycle stages—only the formation extractor changes (rule analyzer → LLM-driven). The formation stage observes 2,262 memory items against 335 for the rule arm (6.8×). Formation recall rises 0.236 (CI $[0.147, 0.331]$) → 0.838 (CI $[0.747, 0.918]$), evolution 0.236→0.802, context 0.695→0.847; first loss is F1_formation in 13/50 instead of 45/50, and 29/50 questions (was 5/50) lose no gold evidence at any stage.

The oracle conditions collapse accordingly. Oracle Memory still repairs every pre-context stage to 1.000 (context 0.979, three context-assembly failures), but its answer-level advantage disappears: 70.8% (34/48) real vs. 71.4% (35/49) oracle memory vs. 70.8% (34/48) oracle context, real-vs-oracle-memory b=4, c=4, p=1.0000, CI $[-0.125, +0.125]$—against +0.170, p=0.0386 under the rule extractor. The failure mix shifts from the evidence side to the evaluation side: F1_formation drops 9→3, E0_evidence_insufficient 7→6, and F6_reasoning (sufficient evidence, full context, wrong answer) appears for the first time, 4 cases, alongside 2 eval_semantic_uncertain and 1 F5_reranking. Formation dominance is a property of the extractor, not of the pipeline: replacing it moves the boundary out of formation and leaves a residual budget dominated by benchmark evidence sufficiency and reasoning.

Two caveats bound this reading. (i) Source coupling: extractor and answer model share a family (qwen2.5:7b), which can only inflate the real-memory condition—the extractor preserves the surface forms the answerer re-uses—so the collapsed oracle gap is, if anything, an underestimate of extractor dependence; the lifecycle-side measurements it rests on are answer-model invariant (Study 2D). (ii) Scale: the residual F6_reasoning budget is not a small-answer-model artifact—at API scale (Study 2B, n=200) F6_reasoning is 3 of 84 attributed failures (3.6%).

Study 3A (external-system replication: Mem0, n=50). The protocol is system-agnostic: an adapter feeds a memory system's artifacts to the frozen attribution code, which matches evidence by content. We re-run the identical protocol—same 50 LongMemEval oracle questions, same answer model and judge (qwen2.5:7b)—on Mem0 (Chhikara et al., 2025), an external open-source system (OSS 2.2.1) whose extraction, embedding and store are independent of our pipeline. Semantic accuracy: Real Memory 58.3% (28/48), Oracle Memory 79.6% (39/49), Oracle Context 77.1% (37/48). Both oracle contrasts survive on the external system (real vs. oracle memory:

b=11, c=1, p=0.0063, CI $[+0.083, +0.333]$; real vs. oracle context: b=11, c=2, p=0.0225, CI $[+0.043, +0.340]$), while the oracle-versus-oracle contrast again does not (p=1.0000, CI $[-0.062, +0.104]$): evidence availability dominates regardless of which system forms the memory.

The failure taxonomy transfers. Mem0 ingests 1,101 messages into 910 extracted facts (283 LLM calls, no update/delete events) against 335 items for our rule extractor on the same questions; its formation recall is correspondingly higher (0.359 vs. 0.236). Yet the answer-failure distribution is nearly identical to the in-house run— F1_formation 9 vs. 9, E0_evidence_insufficient 8 vs. 7, eval_semantic_uncertain 2 vs. 3, F4_retrieval 1 vs. 0, F6_reasoning 2 vs. 0—and formation plus the E0 gate again account for 17 of 22 attributable failures (77%). The gate finds 40 sufficient, 8 insufficient, 2 uncertain; disabling it moves 6 of the 8 insufficient items into F1_formation, inflating the formation count 9→15 (Appendix H). An independent memory stack thus lands on the same boundary structure and the same gate criticality. Stage-recall levels are not comparable across systems (content-tier matching only, and Mem0 ships no reranker, so rerank = retrieval identity); the label distribution is.

6 Analysis

Where memory fails. The stage-recall chain localizes the loss: formation recall is the lowest measured boundary and context recall is high once evidence is stored, so the binding constraint is the extractor, not retrieval. In Study 2B 155/200 real-memory runs lose gold evidence at formation (recall 0.427) while storage/retrieval/reranking sit near ceiling (1.000, 1.000, 0.990); once Oracle Memory injects gold as formed facts the only residual losses are context-assembly (context recall 0.951). The deficit is upstream of retrieval, and its share is a property of the workload: on long, topically diffuse conversations (Study 2C, ∼600 messages) part of the loss migrates downstream (retrieval recall 1.000→0.608, F4_retrieval appears), while formation stays the primary boundary.

The oracle gap is a property of the extractor. The oracle-memory advantage is not a constant of the pipeline family: on the n=50 pilot the rule extractor lets Oracle Memory beat Real Memory by +0.170 (p=0.0386), but under the LLM extractor on the same questions the advantage is +0.000 (p=1.0000, CI $[-0.125, +0.125]$). The mechanism is the stage chain—the rule arm forms 335 items and misses 76.4% of gold evidence, the LLM arm forms 2,262 and misses 16.2%—so once formation recall reaches 0.838 the oracle condition stops discriminating: an oracle ablation on a single extractor measures the extractor, not the architecture.

Reconciling with retrieval-dominant diagnoses. A concurrent LoCoMo diagnostic (Yuan et al., 2026) concludes that retrieval dominates and write policy hardly matters—raw chunk storage with zero LLM calls already beats expensive extraction. The lifecycle view predicts when each conclusion appears: raw-storage-wins is the zero-formation-loss limit, the same null our LLM-extractor arm produces (Study 2F); retrieval-dominates appears when compared retrieval methods span a wide quality range (a 20-point spread in (Yuan et al., 2026)) or conversations grow long enough for retrieval recall to fall (Study 2C: 1.000→0.608, F4_retrieval losses appear); write-dominance appears when extraction is the weakest link (our rule arm loses 76.4% of gold evidence at formation). The dominant boundary is thus a property of write-path quality, retrieval quality, and workload shape; locating it requires measuring every boundary, not pooling one score.

What the E0 gate changes. Before any failure label, the gate asks whether the benchmark's own reference answer is recoverable from the gold evidence; in Study 2B 41/200 items fail this check (and 6 are judge-uncertain), so ≈24% of the nominal error budget is benchmark noise, not memory failure—and as the ablation in Appendix H shows, removing the gate would misattribute those items to formation. Protocols that skip this gate silently convert benchmark noise into system failures; we report E0_evidence_insufficient and eval_semantic_uncertain as first-class outcomes.

The taxonomy transfers across systems. Study 3A runs the frozen protocol on Mem0, an external memory system with an independent extraction stack, on the same 50 questions. Mem0 forms 2.7× more items (910 vs. 335) and reaches higher formation recall (0.359 vs. 0.236), yet its answer-failure distribution matches the in-house run label for label (F1_formation 9 vs. 9, E0_evidence_insufficient 8 vs. 7) and its oracle-versus-oracle contrast is again null. The instrument measures the lifecycle, not our code.

7 Limitations

External validity of the system under test. Formation-boundary numbers characterize this pipeline family, not all memory systems; LLM-native systems with learned extraction (e.g., (Chhikara et al., 2025; Sidik and Rokach, 2026; Long et al., 2026)) may locate their boundary elsewhere. Study 3A addresses this directly: the frozen protocol, run unchanged on an external open-source system (Mem0) with an independent extraction stack, reproduces the failure-label distribution (formation and E0 again 77% of attributable failures) and the oracle contrast structure on the same questions, so the transferable claim is the protocol and the boundary's extractor-dependence (Study 2F), not any single location. The remaining exposure is breadth: one external system on n=50 questions of one benchmark, with provenance-free artifacts (content-tier evidence matching only) and no reranker; a broader system sweep is future work.

Sample size and statistical power. Study 2B runs n=200 (both oracle contrasts $p \leq 0.005$); the controlled ablations 2A/2D/2E/2F use n=10/n=50 subsets and 2C and 3A use n=50, each with paired bootstrap CIs for accuracy differences and stage recall and cluster-robust CIs by conversation (Appendix G). Study 2A (n=10) is a protocol-replication check, underpowered by design. Power simulation for the exact McNemar rule (α=0.05, 20K replicates per point): at n=50 the MDE at 80% power is $\Delta \geq +0.20$ (recovery $\geq$21% against a 2% false-fix rate); the 2B real-vs-oracle-memory contrast (p=0.0386) had 57% power and the 3A contrast (p=0.0063) 85%, and reproducing the 2B effect at 80% power needs $n \approx 80$. Per-type cells (Tables 2 and 3) are descriptive, the smaller ones exploratory.

Judge sensitivity. The judge is an LLM. Study 2E measures cross-family disagreement: prevalence-inflated raw agreement (κ=0.158) with a leniency asymmetry (gemma3:12b over-accepts). Adjudication of all 16 disagreements shifts the n=50 pilot's numbers by ± 4 points without changing ordering; two further blind families sort by leniency rather than by noise (deepseek 13/16 and mistral 1/14 with adjudicated truth; acceptance 40% vs. 90% on the subsample), which bounds, but does not eliminate, adjudicator non-independence. In Study 2B the answer and judge model coincide (deepseek-v4.1-flash); the deterministic rule stack resolves 67/163 real-memory verdicts independently, but the judge-decided remainder inherits this self-judging caveat, and the adjudication covers only the 20-record subsample—all four judges are LLMs, and full human annotation is future work.

Benchmark coverage. LoCoMo categories were mapped from question inspection and secondary docs (table ships with the artifact); benchmarks are English conversational QA with temporal and multi-hop types represented but not exhaustive, and the artifact is anonymized for review.

8 Conclusion

Where does long-term agent memory actually fail? In the families measured, the answer moves with the formation stage: under a rule extractor the boundary sits at formation (155/200 runs lose gold evidence before storage), an LLM extractor relocates it downstream (the oracle advantage collapsing from +17.3 points to zero, Study 2F), and an external system lands on the same attribution structure (Study 3A). That mobility is the transferable contribution: Lifecycle-Boundary Attribution turns a wrong answer into a lifecycle attribution with three oracle controls, an anti-circular gate, an adjudicated judge study, and paired statistics, informative when the boundary moves.

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A Protocol Pseudocode

The attribution decision order is frozen in docs/PROTOCOL.md and implemented once in evaluation/attribution.py. Lexical scores are computed for reporting only and never enter this procedure.

```
def attribute(answer, gold, ref, stages):
    v = verdict(answer, gold)
    if v == CORRECT:                    # (1)
        return NO_FAILURE
    if v == UNCERTAIN:                  # (2)
        return EVAL_SEMANTIC_UNCERTAIN
    if not sufficient(gold, ref):       # (3) E0 gate
        return E0_EVIDENCE_INSUFFICIENT
    for stage in [FORMATION,            # (4) F1-F5
                  STORAGE, EVOLUTION,
                  RETRIEVAL, RERANKING]:
        if not covers(stages[stage], gold):
            return FAILURE_LABEL[stage]
    return F6_REASONING_CANDIDATE       # (5)
```

Frozen thresholds: content-preservation check — shared content vocabulary $\geq 50\%$ of the smaller side and $\geq \min(2, |gold|)$ shared tokens; semantic tier — directional containment ≥ 0.8 with the same absolute floor; deterministic judge support R1 — shared span covers ≥ 0.8 of the candidate (verbatim-quote cases only); deterministic judge support R2 — pure date/relative expressions that resolve to the same date as timestamped evidence. provenance_only matches are reported as diagnostics, never as stage coverage.

B Per-Question Attribution Tables

Every study writes one JSON object per question to comparison.jsonl with: the question and benchmark answer; the gold evidence spans (with ids and timestamps); per-condition answers; the semantic verdicts and judge metadata; the per-stage artifact coverage used by the attribution decision; the assigned failure label; and the code commit of the run. The per-question tables of all studies are released as an artifact alongside the paper; Table 1 and Table 2 aggregate them.

C Judge Prompts and Normalization

The semantic judge is prompted to decide whether the model answer is correct given the gold evidence, and to answer yes/no/uncertain with a one-line rationale; the judge never sees the benchmark reference answer (the E0 gate owns that comparison) and never sees lexical scores. Two deterministic channels repair judge output before it is accepted, both deliberately narrow: R1 lexical accepts a no verdict only when the candidate is essentially a verbatim quote of the evidence (shared span covers ≥ 0.8 of the candidate); R2 temporal accepts pure date/relative expressions that resolve to the same date as timestamped evidence. Anything wider passed wrong answers during calibration ("went to the park" style candidates), which is why the thresholds are frozen and regression-tested. The cross-model judge (Study 2E) uses the same prompt with gemma3:12b; agreement is reported as Cohen's κ on the shared subsample.

D Judge-Study Ledger

Study 2E re-judges a seeded 20-record subsample of the n=50 pilot subset. Every denominator cited in §5 is listed here so the counts are traceable from the per-question artifacts (results/study2e_judge_gemma/) without ambiguity.

Item	Count
Stored answers re-judged (20 questions × 3 conditions)	60
Both judges returned a verdict	55
One judge abstained (eval_semantic_uncertain)	5
Raw agreement, both-verdict	43/55 (78.2%)
Conflicting-verdict pairs	12
Judge-vs-abstention conflicts (of the 5)	4
Adjudicated disagreements (12 + 4)	16
Unadjudicable (preference-type)	2
Decidable cases	14
qwen2.5:7b vs. adjudicated truth	8/14
gemma3:12b vs. adjudicated truth	3/14
Primary-judge verdicts by deterministic rule stack	9/60
DeepSeek re-judges (all 60) vs. qwen / gemma	43/60 / 38/60
DeepSeek vs. adjudicated truth (16 cases)	13/16
Mistral re-judges (all 60) vs. qwen / gemma / deepseek	50/60 / 51/60 / 37/60
Mistral vs. adjudicated truth (14 decidable)	1/14

The 60-answer denominator is the unit of the blind re-judge; the 55/16 split is an analytical view of it, not an additional sample. The 9 deterministic-rule-stack verdicts are primary-judge resolutions made under rules identical for any judge and are excluded from the inter-judge disagreement count. The fourth family (Mistral mistral:7b, local, 60/60 verdicts; three via a payload-echo-safe retry) is lenient like gemma3:12b: it accepts all 41 consensus-core items (41/41) but only 1/14 adjudicated borderline cases. The four families order by leniency—subsample acceptance deepseek 40% < qwen 74% < gemma 90% ≈ mistral 90%—and borderline-band agreement with adjudicated truth follows the same order (13/16 > 8/14 > 3/14 > 1/14), so the adjudicated verdicts track the strictest gold-evidence reading.

E Adjudication Guidelines

The 16 disagreement cases of Study 2E were adjudicated by the authors against the gold evidence and the benchmark reference answer, under the following written criteria, applied identically to every case and blind to which judge produced which verdict:

1. Decidable only on evidence. A case is decidable only if the gold evidence plus the benchmark reference answer determine the verdict. If the reference answer is too vague to grade (two preference-type items), the case is marked unadjudicable and excluded from the strict counts rather than forced.

2. Correct iff fully supported. An answer is correct only if every load-bearing claim it makes is supported by the cited gold evidence and consistent with the reference answer. Concise, paraphrased, and answer-equivalent forms count as correct.

3. No unlicensed inference. An answer that requires a step the evidence does not license (a different entity/event assumed identical, a superlative supported by a single item, a small arithmetic step the reference answer does not itself make) is incorrect. Borderline cases are labeled borderline, not silently forced.

4. Temporal resolution. Relative expressions (yesterday, last Saturday) resolve against evidence timestamps before grading.

5. Independence check. The adjudicated verdicts were then validated against a third judge family (DeepSeek, deepseek-v4.1-flash) re-judging the same answers strictly blind; agreement is reported in §5 and per-case rationales are in the artifact (results/study2e_judge_gemma/adjudication_report.tx

F Stratification Recipes

Study subsets are built by experiments/build_study_set.py. Primary stratum: conversation (or session group) so that no single conversation dominates and within-conversation correlation is spread across the sample; secondary stratum: question type. Allocation is proportional by largest remainder, with a minimum of one question per type present in a conversation. Seed 20261018 for all studies. Committed subsets store question-level records only; conversations are joined at load time from the local full dataset, and raw benchmark text is not redistributed.

G Statistical Procedures

Condition comparisons are paired by question. Exact McNemar: with b questions correct only under condition A and c correct only under condition B, the two-sided exact

p-value is the binomial tail $2\min(P(X \leq \min(b,c)), P(X \geq \max(b,c)))$ with $X \sim \text{Binomial}(b+c, 0.5)$ under the null. Paired bootstrap CI: 10,000 resamples of question indices; percentile interval of the per-question accuracy difference. Cohen's κ (Study 2E): $\kappa = (p_o - p_e)/(1 - p_e)$ over the verdict categories correct/incorrect/uncertain, computed on the seeded subsample of the n=50 pilot subset that both judges re-score. Cluster-robust bootstrap. Because questions within one conversation are correlated, condition CIs are also computed by resampling whole conversations (clusters) rather than questions (paired_stats.py –cluster-by-conversation). LongMemEval oracle-subset questions are sampled one per session (200 questions across 200 distinct conversations, verified), so the naive and cluster CIs coincide there. LoCoMo packs 50 questions into 10 conversations (≈5 each); its cluster-robust CIs are marginally wider than the naive ones (real vs. Oracle Memory: Δ+0.200, naive $[+0.040, +0.360]$ vs. cluster $[+0.038, +0.396]$; real vs. Oracle Context: Δ+0.220, naive $[+0.040, +0.380]$ vs. cluster $[+0.062, +0.408]$) but every positive contrast survives, so the conclusions are unchanged. We report the cluster-robust interval as the conservative bound. Study 3A reuses the n=50 LongMemEval subset (one question per conversation), so its naive and cluster CIs likewise coincide.

Power and MDE. Simulated power of the exact McNemar decision rule (experiments/power_analysis.py; discordant cells drawn from their observed rates, 20K replicates per point, α=0.05 two-sided): at n=50, holding the false-fix rate at the observed 2%, the minimum detectable effect at 80% power is a recovery rate ≥21% ($\Delta \geq +0.19$; $\Delta \geq +0.20$–0.23 across the observed false-fix range). Achieved power at the observed effects: 2B real-vs-oracle memory (b=10, c=2, p=0.0386) 57%, 3A real-vs-oracle memory (b=11, c=1, p=0.0063) 85%, 3A real-vs-oracle context (b=11, c=2, p=0.0225) 67%. Detecting the 2B effect at 80% power would require n≈80; we treat the 2B contrast as significant but modestly powered, and the smaller per-type cells as exploratory.

H E0 Gate Ablation

Reviewer P5 asks what changes if a protocol component is removed. We re-ran attribute_failure on the stored Study 2B (n=200) traces with the anti-circular E0 evidence-sufficiency gate forced sufficient (experiments/e0_gate_ablation.py; no model calls). For real_memory the 10 E0_evidence_insufficient items—answers whose gold evidence cannot answer the question—are re-attributed to a lifecycle stage: 9 become F1_formation and 1 becomes F6_reasoning. The formation-dominant count therefore rises 61→72 (+18%) and the formation share among attributed failures 73%→86%. The gate is what prevents the protocol from conflating question/benchmark limitations with system failures.

The same ablation on the external arm (Study 3A, Mem0, n=50) reproduces the effect on an independent implementation: of the 8 E0_evidence_insufficient items, 6 would be re-attributed to F1_formation (2 to F6_reasoning), inflating the formation count 9→15 (+67%). Gate criticality is a property of the protocol, not of the system under test.

I Environment

Answer/judge models: the n=50 pilot, 2A, 2D, and 2F use qwen2.5:7b (digest 845dbda0ea48, 4.7 GB) and gemma3:12b (digest b92b218582c5, 8.2 GB) served locally via Ollama with temperature 0; Study 2B's n=200 run uses deepseek-v4.1-flash as both answer and judge via the opencode.ai/zen gateway (temperature 0). Retrieval embedding backbone bge-m3:latest (digest 790764642607, 1024-dim). Hardware: single laptop GPU (8 GB VRAM); no distributed inference. Per-study timings and code commits are recorded in each summary.json (models.code_commit); the n=50 pilot and 2A ran under commit a0fbf66 (post-C8 fix), the n=200 Study 2B under commit 84bedce2 (branch fix/formation-diagnostics-evidence-based). Every run is followed by audit_embedding_dims.py to verify the embedding backend served the full run (no silent hash degradation).

Study 3A's external arm runs mem0ai 2.2.1 (MEM0_TELEMETRY=false) with local Ollama qwen2.5:7b extraction, the

bge-m3 embedding backbone, and a persistent Chroma store; conversations are ingested as 4-message chunks, because longer batches return zero memories with local extractors (experiments/run_mem0_external.py, commit 0da190e2).

Responsible NLP Checklist

Datasets. LongMemEval-s (Wu et al., 2025) and LoCoMo (Maharana et al., 2024), both publicly released for research. We use published question-level subsets (LongMemEval-s, LongMemEval oracle subset, LoCoMo) and join conversations at load time from the local full data; raw benchmark text is not redistributed in the artifact (dataset versions and licenses are recorded in the artifact manifest). Both benchmarks contain English conversational dialogues; the subset used here contains no multi-modal records.

Personal data. Both benchmarks are synthetic or licensed fictional dialogues; no personal data of the authors or annotators is used. Speaker names are pseudonyms internal to the benchmarks.

Models and compute. Answer models qwen2.5:7b (Yang et al., 2024) and gemma3:12b, embedding model bge-m3 (Chen et al., 2024), all served locally via Ollama with recorded digests. Total compute is single-GPU (laptop-class, 8 GB VRAM); the full pipeline runs in a few hours per study.

Intended use and misuse. The protocol is intended for diagnosing memory-system failures in research settings. It is not a leaderboard metric: attribution characterizes the concrete implementation family under test (rule-based analyzer plus dense retrieval), and stage-boundary numbers should not be quoted as properties of other architectures. Automated attribution may mislabel cases where gold evidence is ambiguous; we therefore report E0_evidence_insufficient and eval_semantic_uncertain separately instead of forcing a failure label.

Anonymization. The public artifact will be anonymized for review (author-identifying metadata stripped; a separate anonymous branch is prepared by scripts/make_anonymous_branch.sh, with a verification grep and a fresh-repo push recipe for full history anonymization); model-server endpoints and paths are scrubbed.